

Acemoglu, Kong & Ozdaglar (2026)

AI, Human Cognition and Knowledge Collapse — the static problem, Observation 1,
and what Lean says about it

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Repository 4 · Week of September 7 – 12, 2026

`github.com/Arriola123456/ai-04-acemoglu`

Version read: MIT working paper dated May 5, 2026 (NBER WP 34910, issued February 2026);
backup slides after slide 4.

1 · The paper and the agent's problem

Question. Agentic AI gives each person accurate *context-specific* advice. Does it erode the *general* knowledge that makes context-specific information useful — and can a society lose it?

Primitives (Section 3.1). A common state θ_t (random walk; general knowledge) and an idiosyncratic state $\theta_{i,t}$ (own context). Output $f(1\{|x - \theta_t| \leq 1\}, 1\{|y - \theta_{i,t}| \leq 1\})$ with $\Delta_G = f(1, 0) - f(0, 0)$, $\Delta_I = f(0, 1) - f(0, 0)$, $\Delta_X = f(1, 1) - f(1, 0) - f(0, 1) + f(0, 0)$. **Assumption 1:** $\Delta_I = 0$, $\Delta_X > 0$ — context is worthless without general knowledge.

Learning. Effort e costs $\frac{\epsilon}{\epsilon+1}e^{(\epsilon+1)/\epsilon}$ and buys a private signal on own context (precision $\lambda_I e$) and a thin public signal on θ_t pooled into the community's stock (precision $\lambda_G E$, an externality). Agentic AI adds a signal on own context with precision τ_A . Posterior precisions: public X_t ; idiosyncratic $Y = \sigma^{-2} + \lambda_I e + \tau_A$.

The agent's problem (posterior-mean predictions; $G(\tau) = 2\Phi(\sqrt{\tau}) - 1$, $g = G'$):

$$\max_{e \geq 0} U = f(0, 0) + G(X)\Delta_G + G(X)G(Y)\Delta_X - \frac{\epsilon}{\epsilon+1}e^{\frac{\epsilon+1}{\epsilon}}, \quad \frac{\partial U}{\partial e} = \Delta_X G(X)\lambda_I g(Y) - e^{1/\epsilon} = 0.$$

She internalises only her own context; the general knowledge she creates is an externality.

2 · The main result, with all its conditions

Observation 1 (Section 3.4). *Public precision X_t complements human effort, while agentic-AI precision τ_A substitutes for it:*

$$\frac{\partial^2 U}{\partial e \partial X} = \Delta_X \lambda_I g(X) g(Y) > 0, \quad \frac{\partial^2 U}{\partial e \partial \tau_A} = \Delta_X G(X) \lambda_I g'(Y) < 0.$$

- (i) **Assumption 1:** $\Delta_I = 0$ (the private return to effort runs *entirely* through $G(X)G(Y)\Delta_X$) and $\Delta_X > 0$.
- (ii) $\lambda_I > 0$; $X > 0$ so that $G(X) > 0$ (at $X = 0$ effort is worthless: footnote 6).
- (iii) Gaussian facts: $G' = g > 0$ and $g' < 0$ — diminishing returns to precision; $g(\tau) = \varphi(\sqrt{\tau})/\sqrt{\tau}$.
- (iv) Additive precision $Y = \sigma^{-2} + \lambda_I e + \tau_A$: the AI signal and own learning are *perfect substitutes* in producing context-specific precision (Equation 4).
- (v) Posterior-mean predictions (5); the two prediction errors are independent given the information.

Observation 2 (Topkis): $e(X, \tau_A)$ is increasing in X , decreasing in τ_A , strictly for $X > 0$. **Then:** $F(X) = [\Sigma^2 + (X + \lambda_G l e(X, \tau_A))^{-1}]^{-1}$ is strictly increasing (Lemma 1), shifts up in I and down in τ_A (Prop. 2); with $\varepsilon > 4$, above τ_A^c collapse is the only steady state (Prop. 5). **Not claimed:** that AI lowers anyone's context-specific precision — the loss is in the *public* stock X , through effort.

3 · What I did

- ▶ **Version check.** NBER w34910 (Feb 2026) vs the MIT PDF (May 5, 2026, the one I read): same sections and statements; the effort cost is written with a curvature α in February and with the Frisch elasticity ε in May, $\alpha - 1 = 1/\varepsilon$ — so " $\alpha - 1 > 1/4$ " became " $\varepsilon < 4$ ". The issue's "Section 2" is *Related Literature* in both; the model is Section 3.
- ▶ **By hand** (hand/, backup B10): primitives, signals and effort cost (February notation e^α/α), Assumption 1, and Equation (6) under $\Delta_I = 0$.
- ▶ **SymPy** with the real Gaussian: $\partial^2 U / \partial e \partial X > 0$ and $\partial^2 U / \partial e \partial \tau_A < 0$ on 2,000 random parameter points (100%), and $g'(\tau) < 0$ in closed form.
- ▶ **Numerics** (backup B1–B5): best response, the transition map for $\varepsilon = 2$ (unique steady state) and $\varepsilon = 6$ (three, then collapse at $\tau_A^c \approx 0.30$), and steady-state welfare against τ_A : single-peaked at $\tau_A^* \approx 1.0$ ($\varepsilon = 2$), and a cliff to zero at τ_A^c ($\varepsilon = 6$).
- ▶ **Lean via EconCSLib** (lean/, backup B6–B8): 13 source-facing statements of Sections 3.1–3.6 — Equation (6), the FOC as a derivative, both signs of Observation 1, strict decrease, existence and uniqueness of the best response, Observation 2 in X and in τ_A , Lemma 1 and Proposition 2 — all proved, no sorry; lake build: success; check -fast: exit 0. Status *partially formalized*: the Gaussian facts are hypotheses; dynamics and welfare not formalized.
- ▶ **This week's trap** (backup B9): welfare is *not* monotone in AI accuracy (Props. 10–11); Section 5 relaxes aggregation, synthetic data and separability — never the production-side $\Delta_I = 0$.

4 · Where I did not believe the AI

- $f(a,b)$ es la función de utilidad de la tarea.
- El agente trata: $f\{0,1\}^2 \rightarrow \mathbb{R}$, $f(a,b)$ con $f(1,1) - f(0,0) = 1$
- La función de utilidad del agente es: $U_{it} = f(a,b) - \frac{\alpha}{2} \varphi_{it}^2$
- Stocks de conocimiento:
 - $X_{it} := \text{Var}(\theta_{it} | H_{it})^{-1}$ [Stock: Racional]
 - $Y_{it} := \text{Var}(\theta_{it} | H_{it})^{-1}$
- $G(\tau) \equiv \text{Prob. de aceptar con precisión } \tau$
 $G(\tau) := \Pr(|z| \leq \tau) \text{ con } z \sim N(0, \tau^{-1})$

Supuestos:

$$\Delta_G := f(1,0) - f(0,0) \quad \Delta_I := f(1,1) - f(0,0) \xrightarrow{\text{supuesto}}$$

Ganancia asociada al
saber el mundo Ganancia asociada al saber
mi caso

$$\Delta_X := f(1,1) - f(1,0) - f(0,1) + f(0,0) > 0$$

Ganancia asociada más de un caso

Page 2 of the hand derivation: f , U , the stocks X , Y , and Assumption 1 ($\Delta_I = 0$). All three pages in B10.

Claim 1 (an LLM, cold): "welfare rises with AI accuracy — better information cannot hurt." Paper (Section 4.3):

$$\frac{\partial \bar{U}^+}{\partial \tau_A} = \underbrace{g(\bar{Y})G(\bar{X})\Delta_X}_{\text{DE} \geq 0} + \underbrace{\partial_{\tau_A} G(\bar{X}) (\Delta_G + G(\bar{Y})\Delta_X)}_{\text{IE} \leq 0},$$

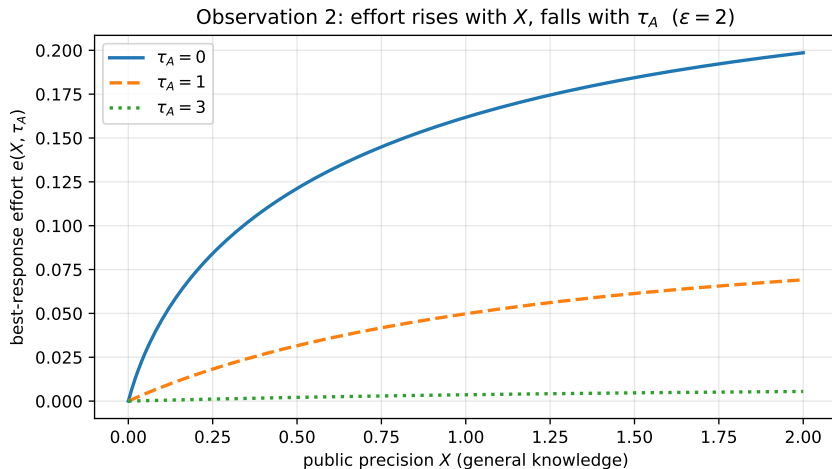
IE/DE rises with τ_A ; under Assumption 2 welfare is single-peaked with finite τ_A^* and $\rightarrow 0$ (Prop. 10); for $\varepsilon > 4$ it drops to 0 at τ_A^c (Prop. 11).

Verdict: wrong — the crowd-out of effort is dynamic and dominates once the AI is good.

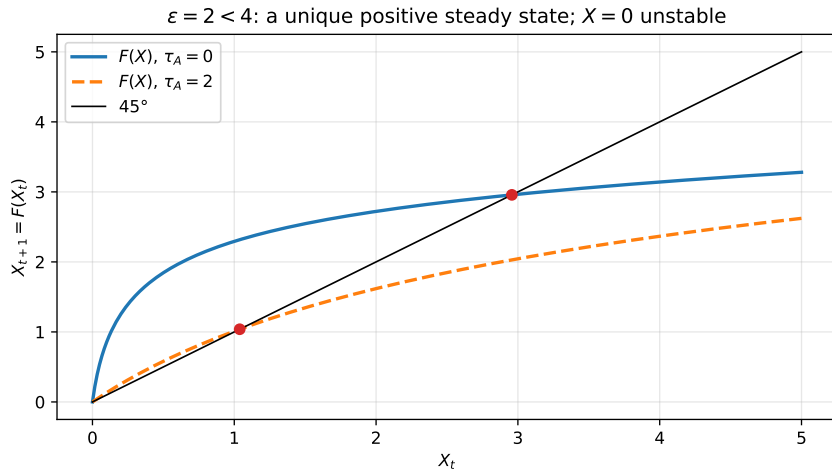
Claim 2 (Lean). Lemma 1 says $F(0) = 0$; Lean's $0^{-1} = 0$ made the formula return Σ^{-2} at $X = 0$. **Verdict: the paper is right as a limit** ($1/0 = +\infty$); the Lean rows live on $X > 0$.

Claim 3 (an LLM on the assumptions). "The authors relax their key assumptions in Section 5." They relax aggregation, synthetic data and separability of effort — all on the *learning* side. The production-side $\Delta_I = 0$ is never touched; with $\Delta_I > 0$ effort does not vanish at $X = 0$ and the collapse fixed point disappears (Lean, B8). **Verdict: incomplete.**

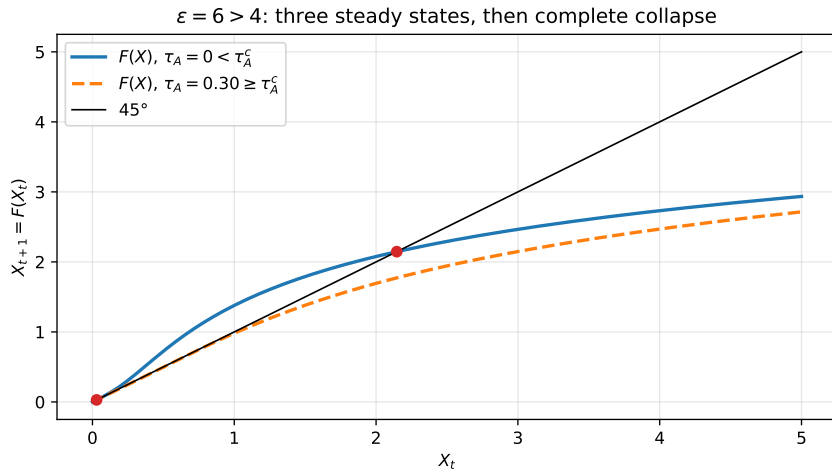
B1 · Best-response effort $e(X, \tau_A)$ — Observation 2



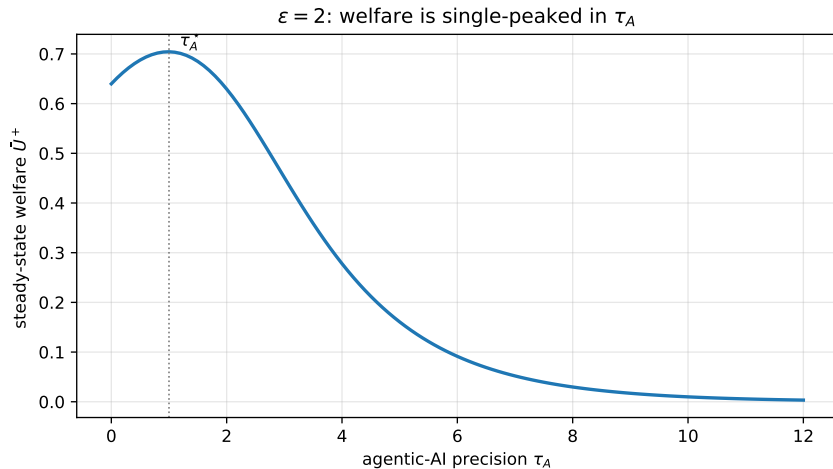
B2 · Transition map F , $\varepsilon = 2 < 4$: a unique positive steady state



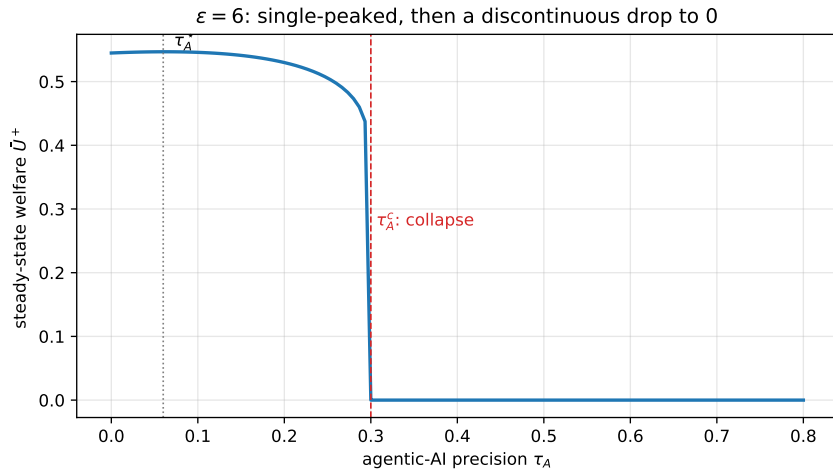
B3 · Transition map F , $\varepsilon = 6 > 4$: three steady states, then complete collapse



B4 · Steady-state welfare against τ_A , $\varepsilon = 2$: single-peaked



B5 · Steady-state welfare against τ_A , $\varepsilon = 6$: a cliff at τ_A^c



Paper claim. $\partial^2 U / \partial e \partial X = \Delta_X \lambda_I g(X) g(Y) > 0$. **Spec** (PaperInterface.lean; G, g abstract with $G' = g, g > 0$):

```
def paper_observation_1_complementSpec : Prop :=
  ∀ (ΔX lamI p0 τ e X ε : ℝ) (G g : ℝ → ℝ),
    0 < ΔX → 0 < lamI → HasDerivAt G (g X) X → 0 < g X → 0 < g (p0 + lamI * e + τ) →
      HasDerivAt (fun X' : ℝ => ΔX * G X' * lamI * g (p0 + lamI * e + τ) - e ^ (1 / ε))
        (ΔX * lamI * g X * g (p0 + lamI * e + τ)) X
        ∧ 0 < ΔX * lamI * g X * g (p0 + lamI * e + τ)
```

Proof (ProofInterface.lean): the chain rule, then the sign.

```
theorem paper_observation_1_complement : paper_observation_1_complementSpec := by
  intro ΔX lamI p0 τ e X ε G g hΔ hlam hG hgX hgY
  refine ⟨?_, ?_⟩
  · have h := (((hG.const_mul ΔX).mul_const lamI).mul_const (g (p0 + lamI * e + τ))).
    sub_const
      (e ^ (1 / ε))
    exact h.congr_deriv (by ring)
  · exact mul_pos (mul_pos (mul_pos hΔ hlam) hgX) hgY
```

Reading. HasDerivAt states the derivative of the marginal utility in X ; mul_pos certifies the sign. The substitute part is identical with $g'(Y) < 0$ and mul_neg_of_pos_of_neg.

FOC. Lean differentiates (6) in e — `Real.hasDerivAt_rpow_const` for the cost, the chain rule for $G(Y)$:

```
def paper_first_order_conditionSpec : Prop :=
  ∀ (f00 ΔG ΔX p0 lamI τ X ε e : ℝ) (G g : ℝ → ℝ),
    0 < e → 0 < ε → HasDerivAt G (g (p0 + lamI * e + τ)) (p0 + lamI * e + τ) →
      HasDerivAt (fun e' : ℝ => f00 + G X * ΔG + G X * G (p0 + lamI * e' + τ) * ΔX
        - ε / (ε + 1) * e' ^ ((ε + 1) / ε))
        (ΔX * G X * lamI * g (p0 + lamI * e + τ) - e ^ (1 / ε)) e
```

Observation 2 in X , stated as a comparison of two FOC solutions (g decreasing and positive on $[\sigma^{-2}, \infty)$):

```
def paper_observation_2_increasing_in_XSpec : Prop :=
  ∀ (ΔX lamI p0 τ ε GX1 GX2 e1 e2 : ℝ) (g : ℝ → ℝ),
    0 < ΔX → 0 < lamI → 0 < ε → 0 < p0 → 0 ≤ τ →
      AntitoneOn g (Set.Ici p0) → (∀ y ∈ Set.Ici p0, 0 < g y) →
      0 ≤ GX1 → GX1 < GX2 → 0 ≤ e1 → 0 ≤ e2 →
      ΔX * GX1 * lamI * g (p0 + lamI * e1 + τ) = e1 ^ (1 / ε) →
      ΔX * GX2 * lamI * g (p0 + lamI * e2 + τ) = e2 ^ (1 / ε) →
      e1 < e2
```

Proof idea. If $e_2 \leq e_1$ the benefit at (e_2, X_2) exceeds $e_1^{1/\varepsilon} \geq e_2^{1/\varepsilon}$: contradiction.

One lemma carries Lemma 1 and Proposition 2: $A \mapsto (\Sigma^2 + A^{-1})^{-1}$ is strictly increasing on positives.

```
def paper_lemma_1_transition_strictly_increasingSpec : Prop :=
  ∀ (Sig2 lamG I : ℝ) (e : ℝ → ℝ),
    0 < Sig2 → 0 < lamG → 0 < I → MonotoneOn e (Set.Ici 0) → (∀ X, 0 ≤ e X) →
      StrictMonoOn (fun X : ℝ => (Sig2 + (X + lamG * I * e X)-1)-1) (Set.Ioi 0)
```

Boundary. Lean's $0^{-1} = 0$: the paper's $F(0) = 0$ is a limit convention, so the rows live on $X > 0$:

```
theorem transition_at_zero_lean_convention (Sig2 lamG I : ℝ) :
  (Sig2 + ((0 : ℝ) + lamG * I * 0)-1)-1 = Sig2-1 := by simp
```

The untouched assumption. With $\Delta_I > 0$ the marginal benefit of effort at $X = 0$ is positive (and, by transition_positive_with_effort, $F(0) > 0$):

```
theorem effort_incentive_survives_without_general_knowledge (ΔI ΔX lamI p0 τ : ℝ) (g : ℝ → ℝ)
  (hΔI : 0 < ΔI) (hlam : 0 < lamI) (hg : 0 < g (p0 + τ)) :
  0 < ΔI * lamI * g (p0 + τ) + ΔX * 0 * lamI * g (p0 + τ)
```

What Lean verifies. 13/13 rows closed; lake build: success (4000 jobs); check -fast: exit 0. Not derived: the Gaussian facts $G' = g$, $g' < 0$ — hypotheses, the declared boundary.

Is welfare increasing in AI accuracy? No.

- ▶ $\varepsilon < 4$ + Assumption 2: $\bar{U}^+$ strictly increasing on $(0, \tau_A^*)$, strictly decreasing after, $\rightarrow 0$ (Prop. 10).
- ▶ $\varepsilon > 4$: same up to τ_A^c , then $\bar{U}^+ = 0$ (Prop. 11); also the basin threshold $\bar{X}_m$ rises with τ_A (Prop. 7).
- ▶ Scaling law: $\tau_A^c / \log I \rightarrow 2/\varepsilon$, $\tau_A^* / \log I \rightarrow 2/(\varepsilon + 1)$ (Prop. 12); without Assumption 2 the profile need not be single-peaked (footnote 8).

Why the intuition fails: effort is under-supplied (externality); a better AI lowers effort (Obs. 2), which lowers $\bar{X}_h$ (Prop. 2, 4, 8); the indirect loss grows with τ_A while the direct gain has diminishing returns (g falls).

Assumptions the authors relax (Section 5).

-
- | | |
|--------------------|---|
| 5.1 Aggregation | $I(\tau_A) = I_0 + e^{\eta\tau_A}$; results survive if $\eta < \varepsilon/2$ (Prop. 14) |
| 5.2 Synthetic data | $\tau_{syn} > 0$ puts a floor under X (Prop. 15) |
| 5.3 Separability | public signal $\propto e^\beta$; threshold $\varepsilon < 4/\beta$ (Prop. 16) |
-

Never touched: Assumption 1's $\Delta_I = 0$ — context-specific knowledge has zero standalone value. It is a production-function assumption (page 9), and it is what makes the private return to effort vanish with X , what makes $(\bar{X}, \bar{e}) = (0, 0)$ a steady state, and what gives the collapse state welfare 0 (Section 4.1). With $\Delta_I > 0$: effort at $X = 0$ is $\Delta_I \lambda_I g(\sigma^{-2} + \tau_A) > 0$, $F(0) > 0$, no collapse fixed point (Lean, B8). Also untouched: the additive precision in (4), i.e. AI and own learning as perfect substitutes for context.

B10 · Hand derivation: primitives, Assumption 1, Equation (6)

Acemoglu, Koy & Ozdaglar (2026):

- θ_t es el conocimiento ~~del agente~~, con $\theta_{t+1} = \theta_t + \varepsilon_t$
con $\varepsilon_t \sim N(0, \Sigma^2) \Rightarrow \theta_{t+1} \sim N(0, \sigma^2)$.

- El agente escoge cuanto esforzarse:

$$\text{costo} = \frac{1}{\alpha} e^{\alpha} \geq 0, \quad \alpha > 1$$

Para esto compra 2 variables:

$$S_{it}^H \sim N(\theta_{it}, (\lambda_{it} E_{it})^{-1}) \quad [\text{Privada}]$$

$$S_{it}^C \sim N(\theta_{it}, (\lambda_{it} E_{it})^{-1}), \quad E_{it} = \int_{I_{it}} e_{it} di \quad [\text{Pública en la Isla}]$$

Es una estimación, por lo que el agente ignora su aporte a E_{it} .

- La IA agnition tiene: $S_{it}^A \sim N(\theta_{it}, \tau_A^{-1})$
antes de la IA se tiene que $\tau_A = 0$.

- Se conoce X_{it} sobre el estado común θ_t , ~~pero~~ sobre θ_{it} no tiene Y_{it} . (obra según variables de estado indicadoras.)

$$a := \mathbb{1}\{|x_{it} - \theta_{it}| \leq 1\} \quad b := \mathbb{1}\{|\eta_{it} - \theta_{it}| \leq 1\}$$

- $f(a, b)$ es la función de producción de la tarea.

- El agente tiene:

$$f: [0, 1]^2 \rightarrow \mathbb{R}, \quad f(a, b) \text{ con } f(1, 1) - f(0, 0) = 1$$

- La función de utilidad del agente es:

$$U_{it} = f(a, b) - \frac{1}{\alpha} e_{it}^{\alpha}$$

- Stacks de conocimiento:

$$\circ X_{it} := \text{Var}(\theta_{it} | H_{it})^{-1} \quad [\text{Stack Privada}]$$

$$\circ Y_{it} := \text{Var}(\theta_{it} | H_{it})^{-1}$$

- $G(\tau) \equiv \text{Prob. de aceptar con precisión } \tau$

$$G(\tau) := \Pr(|Z| \leq \tau) \text{ con } Z \sim N(0, \tau^{-2})$$

Suposiciones:

$$\Delta_G := f(1, 0) - f(0, 0) \quad \Delta_X := f(1, 1) - f(0, 1) = 0 \quad \text{Sumando}$$

Ganancia aceptando solo sobre el mundo Ganancia aceptando solo sobre mi caso

$$\Delta_X := f(1, 1) - f(1, 0) - f(0, 1) + f(0, 0) > 0$$

Ganancia aceptando más de un caso

$$Y_{it} = \sigma^{-2} + \lambda_{it} e_{it} + \tau_A, \text{ considerando esto}$$

$$M_{it} = \frac{\lambda_{it} e_{it}^{\alpha} + \tau_A S^A}{Y_{it}}$$

$$U_{it} = f(a, 0) + G(X_{it}) \Delta_G + G(Y_{it}) G(X_{it}) \Delta_X - \frac{1}{\alpha} e_{it}^{\alpha}$$

hand/hand-derivation.pdf, February notation: cost e^{α}/α , $\alpha - 1 = 1/\varepsilon$ in the May version. Page 3 boxes Equation (6).